Triple Scissor Extender Robot Arm: A Solution to the Last One Foot Problem of Manipulation

Abbas Shikari and Harry Asada, *Member, IEEE*

***Abstract*—Final assembly, inspection, and maintenance of aircrafts, automobiles, and industrial equipment often requires a robot to reach confined areas that are difficult to access. While a robot can move freely in open space, the last one foot to a target site in a manufacturing environment is typically cluttered with complex obstacles. Here, we present a novel solution to the last one foot problem. Inspired by plant stem growth, a new design concept for an expandable arm, the triple scissor extender robot arm (TSERA), is developed to access a confined area through a narrow channel. The robot consists of a series of expandable segments each of which is able to extend and tilt. Thus, the robot can elongate its structure along a narrow, winding space. The novel design concept is accompanied by a new algorithm for path planning, the sequential expansion algorithm. Unlike traditional algorithms for articulated robot path planning, TSERA's sequential expansion algorithm reduces the otherwise computationally expensive multibody path planning problem to a single rigid body kinematics problem. Exploiting the serial nature of expandable structure, we can compute arm motion by sequentially solving a simple inverse kinematics problem for each segment.**

***Index Terms*—Industrial robots, kinematics, parallel robots, redundant robots.**

I. Introduction

INDUSTRY workers are frequently tasked with reaching target points through narrow gaps between obstacles during the assembly, maintenance, and inspection of complex machinery. The aircraft and automobile manufacturing industries, for example, rely on their technicians ability to fit into tight spaces for visual, force, and torque inspection along with various fastening tasks. Due to human kinematic and spatial limitations, however, these tasks can result in worker injury and decreased productivity. These tasks are laborious and fatiguing, yet no effective technology is currently available for replacing the manual labor, or assisting the workers to reduce their workload. Except for the development of problem-specific, single-purpose solutions, no generic technology has been established and applied to practical tasks.

Traditional robotic arms are ill fitted to accessing confined areas. The kinematic and structural properties of traditional articulated robots are a mismatch for tasks involving accessing confined spaces. Bulky shoulder, elbow,and wrist links may interfere with obstacles at the last one foot area, and are size excluded within narrow channels. Additionally, open kinematic chain robots must solve complex path planning problems when accessing confined spaces. These robots must perform computationally expensive collision checks for a full trajectory of every rigid body of the robot in order to generate a feasible path.

Issues pertinent to confined space inspection and manipulation have been addressed in three major areas in robotics literature. Researchers in the field of soft robotics have produced several novel designs in recent years. These include parallel pneumatic bellows [1], pressure driven eversion [2], [3], and series pneumatic artificial muscle [4] robots. These systems offer a high degree of flexibility over rigid robotic systems in tasks such as search and rescue [5]. However, their small payload capacity and low positioning accuracy limit their utility in manufacturing environments.

Unlike soft robots, snake and endoscope-like robots have made significant strides in entering manufacturing environments. Specifically, the company OC Robotics has created several tethered robotic systems capable of industrial inspection and various machining operations within a confined space [6]. Despite their success, OC Robotics tethered systems require a hefty set of actuators at the base of the system that could limit feasibility in cases like aircraft assembly in which significant infrastructure would need to be arranged for a large base to access confined spaces within the hull of the aircraft. Notable research efforts have also been made in the fields of mobile snake robots [7]–[9] and endoscopic robots [10], [11] capable of highly dexterous motion via environmental interaction. While these mobile systems are well suited for certain tasks, they are limited in a variety of use cases. These include environments in which friction driven locomotion is not allowed. In more restrictive cases, contact with the environment is prohibited. In other cases, snake and endoscope-like robots are difficult to apply due to the lack of supporting structure in environments composed of irregular confined spaces. Furthermore, bringing the snake robot to the channel inlet is difficult without significant infrastructure.

This letter presents a new approach to accessing confined spaces in manufacturing environments. The new design, called the Triple Scissor Extender Robot Arm, TSERA, is a self sup-

Manuscript received February 24, 2018; accepted July 4, 2018. Date of publication July 24, 2018; date of current version August 8, 2018. This letter was recommended for publication by Associate Editor C. Gosselin and Editor P. Rocco upon evaluation of the reviewers' comments. This work was supported by NSK, Ltd. *(Corresponding author: Abbas Shikari.)*

The authors are with the d'Arbeloff Laboratory for Information Systems and Technology, Department of Mechanical Engineering, Massachusetts Institute of Technology, Cambridge, MA 02139 USA (e-mail: ashikari@mit.edu; asada@mit.edu).

This letter has supplementary downloadable material available at http://ieeexplore.ieee.org, provided by the authors. The Supplementary Materials contain a video demonstrating the workspace and mobility of the prototype TSERA within a confined space.

Digital Object Identifier 10.1109/LRA.2018.2859022

2377-3766 © 2018 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

Authorized licensed use limited to: Worcester Polytechnic Institute. Downloaded on September 02,2026 at 14:14:37 UTC from IEEE Xplore. Restrictions apply.



Fig. 1. Expandable Robot Arm in a narrow channel where traditional robot arms are unable to access.

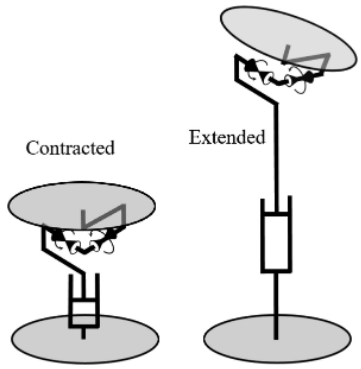


Fig. 2. Single segment of expandable arm consisting of one prismatic and two revolutionary joints.

porting robotic arm capable of moving through narrow, winding spaces while carrying a significant payload. As shown in Fig. 1, the robot consists of several segments connected end to end. Each segment can exhibit three-DOF: elongation along the central axis and two axes of tilting motion. The structure has high stiffness and high positioning accuracy relative to its base frame. Additionally, a computationally efficient algorithm, the Sequential Expansion Algorithm, is presented to allow the arm to traverse winding trajectories through narrow channels with minimal spill out and accurate endpoint positioning. Unlike traditional motion planning algorithms in which numerous collision checks must be performed for full trajectories of every rigid body before an interference-free motion can be made, TSERA's path planning algorithm only requires collision checks for a single rigid body before guaranteeing collision free motion.

In the following sections, the basic architecture and principle of the Triple Scissor Extender Arm are presented, followed by mechanical design, kinematic, and static analysis, and motion planning algorithms. Lastly, a prototype is designed, built, and tested to demonstrate the efficacy of the design and the algorithm.

II. Sequential Expansion

A plant extends its stem at its growing point. It adds a new segment to the tip and leaves the segment behind. This allows a plant to extend its stem through even a narrow gap. Inspired by this plant growth, we present a Sequentially Expandable Arm that can expand a series of expandable segments. This mechanism allows for extending the arm through a narrow gap and reaching a point, which is otherwise difficult to access.

Unlike the plant, however, mechanical robots are unable to transport materials to the tip and convert them to a structure. Instead, an engineered robot can consist of several expandable segments serially connected. Beginning from a fully contracted position, the robot can expand each of its segments in sequence in order to reach a desired point. This can be made possible if each segment possesses a prismatic degree of freedom to expand the segment and rotational degrees of freedom to tilt the segment in two directions, e.g. pan and tilt. Fig. 2 shows the schematic of such a segment.

By stacking a number of these expandable segments end-to-end, we can create a robot arm that mimics plant stem growth. As shown in Fig. 3-1, all the segments are initially contracted, and each segment expands sequentially starting with the proximal segment. The sequence shown in Fig. 3 illustrates this Sequential Expansion Algorithm. Unlike the plant stem growth, this robotic system is reversible, so that the arm can retract after completing a task at a destination. Therefore, it can serve for practical tasks in manufacturing and maintenance.

The Expandable arm consists of N segments with the proximal segment being Segment 1. The key is to expand the proximal segment first, followed by the second segment. When expanding Segment 1, all the segments, 2 through N, are fully contracted and are treated as a single rigid body together with the end-effector. This single rigid body is called Body $2-N$. A trajectory can be found for the short interval from the initial starting point to a waypoint, which is a point along the centerline of the open space envelope that Segment 1 can reach. Possible interference between Body $2-N$ and the envelope must be checked in connecting points. After connecting the first two waypoints, Segment 1 is fixed, and Segment 2 alone is released. This procedure is repeated for each of the distal segments, 3 through N. Note that Body $k-N$ becomes smaller as this procedure repeats, i.e. k increases. The algorithm is terminated when it reaches the target point or all the segments have been expanded.

As illustrated in Fig. 3, the Serial Expansion Algorithm allows the expandable robot arm, such as TSERA, to go through a narrow channel, which is infeasible or difficult for conventional multi-link robots to pass through. Furthermore, this algorithm provides us with an effective method for motion/path planning. Compared to conventional path planning for articulated robots, this procedure is much simpler. In each step the only interference check required between motions is between a single rigid body and a fixed envelope. This is juxtaposed with the path planning algorithms of traditional robots which require checking for interference for numerous rigid bodies over a full time series of motions before making a single move within a confined space [12], [13]. This Sequential Expansion Algorithm solves the path planning problem by dissolving it into a sequence of simple, small-scale planning problems.

After reaching a target point, the dexterity and workspace of the expandable arm is determined by the number of unexpanded segments. Thus, when selecting an optimal set of waypoints along a given trajectory, the path planner should place waypoints such that the proximal segments of the arm are as close to maximum extension as possible while avoiding obstacle collisions.

Authorized licensed use limited to: Worcester Polytechnic Institute. Downloaded on September 02,2026 at 14:14:37 UTC from IEEE Xplore. Restrictions apply.

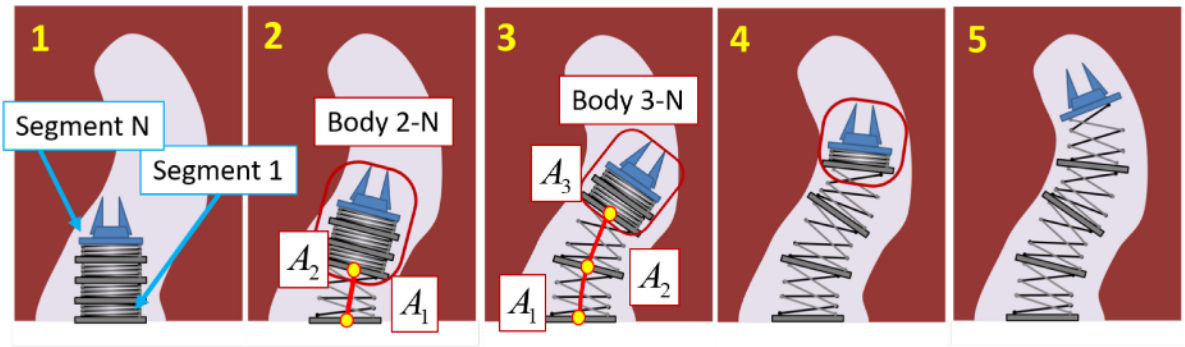


Fig. 3. Depiction of the Sequential Expansion Algorithm: A series of segments expands in sequence starting from the most proximal segment towards the most distal segment following a narrow channel.

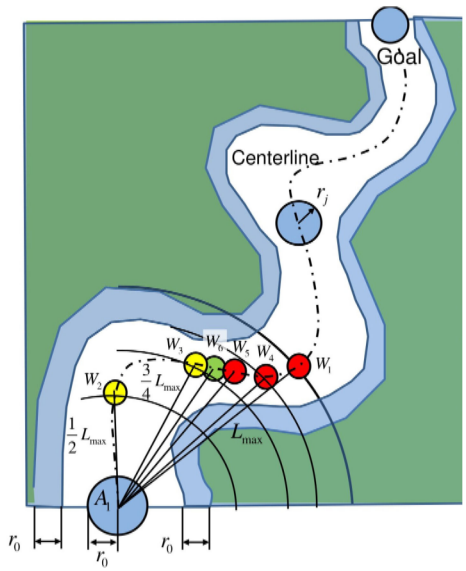


Fig. 4. Optimal waypoint placement for the Sequential Expansion Algorithm. Each waypoint is placed within the reach of the segment on the centerline that is furthest from the previous waypoint but the line between the consecutive waypoints does not interfere with the the narrow channel.

The algorithm for optimal placement of waypoints is depicted in Fig. 4. Assuming the obstacle map is known, the waypoints are determined via a binary search algorithm. In the figure the prohibited area is shown in dark green. First assume that the robot is fully contracted and located at position A_1. Possible collision between the prohibited area and Body $2-N$, the rigid body of the first step of expansion, must be checked. Using the Configuration Space Approach, the available space for expansion without collision can be determined, as shown by the white area in the figure [13]. For simplicity the rigid body is represented as a ball of radius r_0 in the figure. With this method Body $2-N$ reduces to a volume-less point, while the available space reduces. For this collision-free area, the centerline is determined as shown by a dot-and-dash line.

Let L_{max} represent the maximum extended length of a segment of the expandable arm. Point W_1 is a candidate waypoint on the centerline at a distance L_{max} from point A_1. The line between points A_1 and W_1 intersects with the wall of the collision-free area. Therefore, this test point W_1 is rejected. Next, the radius of the circle is halved: $L_{max}/2$, and the second waypoint candidate W_2 is determined, as shown in the figure. This time the line between A_1 and W_2 lies entirely within the collision-free area. Next, the radius is set to 3/4 of the original length and the same procedure is repeated. In the figure, yellow waypoint candidates are collision-free, while red waypoint candidates interfere with the wall. Continuing this binary search, it arrives at the green waypoint W_6 that is the furthest point on the centerline from the starting point. Point W_6 is redefined as point A_2. The first segment is to be extended to this waypoint, and the same procedure is repeated to the next segment. It is stopped when arriving at the goal, or all the segments have been extended.

Once all the waypoints are determined, the prismatic joint displacement and the two tilting angles of each segment can be determined. Namely, the prismatic joint displacement of the $i-th$ segment is simply the distance between A_i and A_{i+1}, and the tilting angle is determined by the direction of the straight line from A_{i+1} to A_{i+2}. The orientation of the final segment determines the orientation of the end-effector attached to it. The orientation is not determined by the next waypoint, but by the desired orientation of the end-effector.

The sequential expansion algorithm with binary waypoint search is easily implemented for an ideal structure as depicted in Fig. 2. The algorithm can be applied to a broader class of expandable robots where individual segments can be extended and tilted in sequence. The following section shows a particular case where the algorithm is applied with minor modification.

III. Design

An expandable arm must meet a set of kinematic, structural, and geometric design criteria in order to enable the sequential expansion algorithm and successfully access confined spaces within manufacturing environments. Functional requirements and metrics for design evaluation include:

- The arm must be composed of several segments stacked in sequence. Each segment should be capable of three-DOF motion, extension, and two-DOF tilt. Fig. 2 shows the desired kinematics of a single segment.
- To allow high positioning accuracy the structure should maintain high axial, torsional, and bending stiffness.

Authorized licensed use limited to: Worcester Polytechnic Institute. Downloaded on September 02,2026 at 14:14:37 UTC from IEEE Xplore. Restrictions apply.

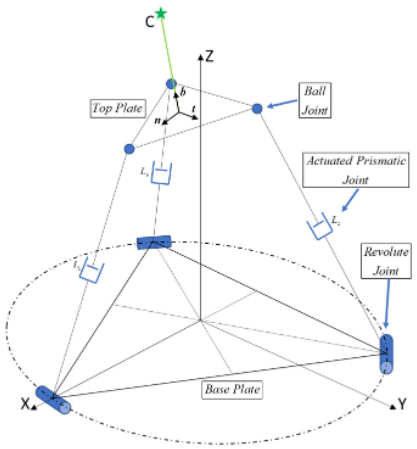


Fig. 5. Kinematic structure of a single segment of TSERA consisting of three prismatic joints.

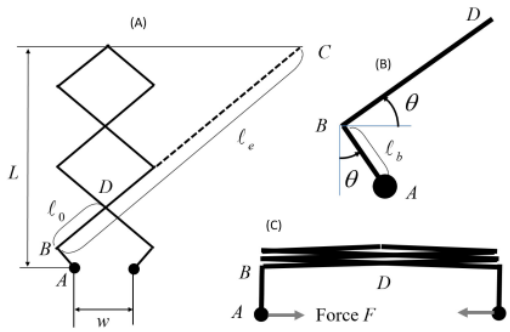


Fig. 6. Kinematic diagram of a single scissor linkage, (A) The height L is a function of the actuator displacement w. Length l_e represents the length $\overline{BC}$, (B) The bottom link of the scissor mechanism is bent 90 degrees to prevent a singular configuration from occurring, (C) The standard scissor mechanism becomes singular when it is completely contracted. The offset l_b can eliminate the singularity.

- The arm must be of low profile when fully contracted, while the body must be expanded to reach a point deep in a confined area. The ratio of the fully expanded body length to the fully contracted body length, termed expansion ratio, is the figure of merit for this criteria.
- The cross-sectional area of the arm must be small relative to the fully extended length of the structure to minimize possible interference with the environment. To evaluate the slenderness of the arm we use the ratio of the maximum diameter of the cross-sectional area to the maximum length of the structure.

Parallel mechanisms exhibit many of the characteristics that meet the design requirements of a segment of the expandable arm. These include high load bearing and stiffness [14]. However, these structures are typically extremely limited in their workspace [15]. The Stewart mechanism, for example, has a small work space. Gonzalez and Asada report on a parallel structure with enlarged workspace along the central axis [16]. This structure's workspace is enlarged with the addition of scissor linkage mechanisms in place of traditional prismatic joints [17]. See Fig. 6. Although the extension ratio is high (on the order of 5 to 10), the large footprint of its base frame due to the placement of the six actuators diminishes the applicability of their original design to the last one foot problem.

Because extension and two tilting axes are the only required DOF for each segment of the expandable arm, the mechanism can be simplified and made more compact. We selected a structure belonging to the family of zero-torsion parallel mechanisms for each of the arm segments [18], [19]. This family of mechanisms is characterized by a mobile platform that is connected to three legs via spherical joints. These legs constrain the spherical joints to move within three equally spaced vertical planes that intersect along a common line. The kinematics, workspace, and singularities of this class of mechanisms have been extensively studied by Bonev and Briot [18], [19].

Fig. 5 shows the basic kinematic structure of a single segment of the expandable arm. The top and bottom plates are connected with three prismatic joints driven by independent actuators. The top end of each prismatic joint is a spherical joint. The bottom end of the prismatic joint is a revolute joint. When the three prismatic joints lengthen an equal distance, the top plate is lifted and the structure elongates. A difference in length among the three prismatic joints causes the top plate to rotate about any axis perpendicular to vertical.

One of the major problems with scissor mechanisms is singularity. As shown in Fig. 6-(C), the scissor mechanism becomes singular when it is completely contracted. A horizontal force applied to the end of the bottom link, point B, does not generate an upward motion at this singular configuration. Furthermore, a large actuator force is required near the singular configuration. However, this singularity problem can be alleviated by extending the bottom end in the direction perpendicular to the bottom link; a horizontal force applied to point A can generate a moment about point D with a moment arm l_b. In our design, a linear actuator is connected to point A to alleviate the singularity problem.

During the review process of this letter, a few alternative approaches to expandable robot design have been found. The zipper robot [20], [21] has an extremely high expansion ratio 14:1 with a novel mechanism, but is difficult to connect multiple segments in series. The pneumatic pressure-driven eversion robot [2] can extend its tip with a novel pneumatic tube structure and allows for steering its tip in a highly cluttered environment. However, it is difficult to carry an end-effector at the tip, and is unable to locate the end-effector at an arbitrary location.

IV. Analysis

A. 2D Scissor Linkage

We begin with the determination of the relationship between scissor linkage height and the actuator displacement for a single scissor mechanism. This is a planar linkage system, as shown in Fig. 6. Let L be the height of the scissor and w be the distance between end points of base scissor links, which is varied by an actuator.

A salient feature of this scissor mechanism is the offset in each of the lowest linkages. The length of the offset is denoted l_b in Fig. 6. The singularity that occurs at the completely contracted configuration (C) can be eliminated, and the peak motor torque, which is required at the contracted configurations, is reduced by using this offset, at the cost of complicating the scissor linkage kinematics.

Authorized licensed use limited to: Worcester Polytechnic Institute. Downloaded on September 02,2026 at 14:14:37 UTC from IEEE Xplore. Restrictions apply.

Since all links are kept parallel, the height of the scissor linkage is given by

$$L = l_e \sin\theta + l_b \cos\theta \tag{1}$$

where l_e is the constant length of line $\overline{BC}$ and θ is the inclination angle of l_e to the horizontal. The width of the scissor linkage is similarly related to another non-material line, $\overline{AD}$ with length l_p as follows,

$$w = 2l_0 \cos\theta - l_b \sin\theta \tag{2}$$

From (1) and (2) we find

$$\cos\theta = \frac{Ll_b + \frac{w}{2}l_e}{l_b^2 + l_0 + l_e} \quad \sin\theta = \frac{Ll_0 - \frac{w}{2}l_b}{l_e l_0 + l_b^2} \tag{3}$$

Using the trigonometric identity $\sin^2\theta + \cos^2\theta = 1$, we can obtain

$$w = \frac{2}{l_b^2 + l_e^2}\left[(l_b^2 + l_o l_e)\sqrt{l_b^2 + l_e^2 - L^2} - Ll_b(l_e - l_o)\right] \tag{4}$$

Using these kinematic relationships, let us obtain the actuator force F required for generating a force Mg, or bearing a load $-Mg$, at the tip of the scissor mechanism. Assuming that the mechanism is friction-less and mass-less, the virtual work done by the actuator force and the external force is given by

$$\delta Work = -F\delta w - Mg\delta L \tag{5}$$

Limiting the range of θ to $0 < \theta < \pi/2$, virtual displacements δL and δw are determined by $\delta\theta$. From (1) and (2),

$$\begin{aligned}\delta L &= (l_e \cos\theta - l_b \sin\theta)\delta\theta \\ \delta w &= -2(l_0 \sin\theta + l_b \cos\theta)\delta\theta\end{aligned} \tag{6}$$

Substituting (6) into (5), we can find that the virtual work vanishes for an arbitrary $\delta\theta$ if, and only if,

$$F = \frac{Mg(l_e \cos\theta - l_b \sin\theta)}{2(l_0 \sin\theta + l_b \cos\theta)} \tag{7}$$

Note that for $0 < \theta < \pi/2$, the actuator force is finite and decreases as the offset l_b increases. Note also that (7) is a monotonically decreasing function of θ. Therefore, a larger actuator force is required as it is getting closer to the singular configuration. But, it is bounded even in the vicinity of the singular point.

$$F < Mg\frac{l_e}{2l_b} \tag{8}$$

The offset l_b reduces the upper bound of the actuator force. This technique does not eliminate the internal structural singularity at $\theta = 0$, but the required actuator force is effectively reduced.

B. Segment Kinematics and Parasitic Displacement

Although the zero-torsion parallel mechanism is able to produce the desired motion for a single segment of the expandable arm, there is coupling between tilting motions and $xy-$ translation. Tilting the top plate inevitably induces a sideway displacement. The workspace of the zero-torsion parallel mechanism has been analyzed by Bonev in [18]. In his letter, Bonev found that the magnitude of the sideways, parasitic translation, here denoted r, was entirely dependent upon the angle of tilt of the top plate, γ,

$$r = \frac{1}{2}R_t(1 - \cos\gamma) \tag{9}$$

where R_t is the radius of the circumcircle defined by the vertices, E_i, $i \in (1, 2, 3)$.

The parasitic motion must be accounted for in our motion planning algorithm in order to avoid additive positioning error of the end point. The following analysis is to evaluate the sideway deflection in relation to the top plate tilting angle, and find a series of segment configurations, which are modified by taking into account the sideway deflection. First, we analyze a single scissor mechanism, find the functional relationship between the top plate tilting angle and the parasitic translation, and then obtain the modified segment configuration to compensate for the sideway deviation for the series of segments.

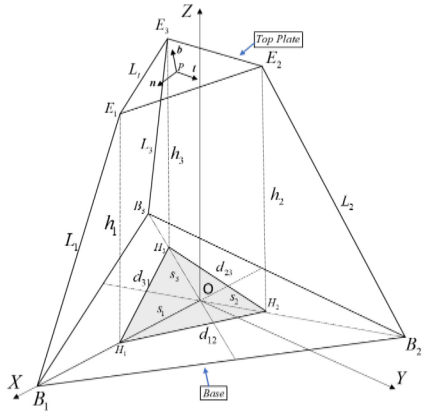


Fig. 7. Coordinates and variables used to characterize the kinematics of a single segment.

The coordinate frame of the top plate of a single segment of TSERA is shown in Fig. 7. The top plate position is given by $P : \mathbf{p} = [x, y, z]^T$ in the O reference frame . The orientation of the top plate is captured in the unit vectors, $\mathbf{n}$, $\mathbf{t}$, and $\mathbf{b}$. As noted in the previous chapter, the orientation of the top plate is coupled with translational motion along the X and Y axes. Thus, the unit vectors, $\mathbf{n}$, $\mathbf{t}$, and $\mathbf{b}$ can be expressed as functions of x and y . The goal for this section is to relate height of the three scissor linkages, L_i to end effector position, P .

To simplify the following analysis, we introduce a set of intermediate variables. s_i is the distance between the origin, O, and point H_i, the projection of the top plate vertex, E_i, onto the base plate. Here the index, $i \in (1, 2, 3)$, indicates any of the three vertices of the base plate. h_i is the distance between H_i and E_i. d_{ij} is the length of the projection of the top plate onto the base . Lastly, s_0 is a constant characteristic length of the triangle defined as the distance from the center, O to any of the base vertices, B_i . From these definitions, it follows that each scissor length, is given by,

$$L_i = \sqrt{(s_0 - s_i)^2 + {h_i}^2} \tag{10}$$

Authorized licensed use limited to: Worcester Polytechnic Institute. Downloaded on September 02,2026 at 14:14:37 UTC from IEEE Xplore. Restrictions apply.

We now find a relationship between the internal variables and position coordinates of the top plate. Consider, $\triangle H_1OH_2$ in Fig. 7. Note $\angle H_1OH_2 = \frac{2\pi}{3}$ since O is located at the center of the base. From the law of cosines it follows that,

$$ {d_{12}}^2 = {s_1}^2 + {s_2}^2 + s_1 s_2 \tag{11} $$

The projected dimension, d_{12}, can be related to the top plate via the following equation:

$$ {d_{12}}^2 + (h_2 - h_1)^2 = {L_t}^2 \tag{12} $$

Substituting (11) into (12) gives the first of three kinematic constraint equations,

$$ {s_1}^2 + {s_2}^2 + s_1 s_2 + (h_2 - h_1)^2 = {L_t}^2 \tag{13} $$

Two additional constraint equations are found by following the same procedure,

$$ \begin{aligned} {s_1}^2 + {s_3}^2 + s_1 s_3 + (h_3 - h_1)^2 &= {L_t}^2 \\ {s_3}^2 + {s_2}^2 + s_3 s_2 + (h_2 - h_3)^2 &= {L_t}^2. \end{aligned} \tag{14} $$

The coordinates of the vertices, E_1, E_2, and E_3, are expressed in terms of the internal variables:

$$ \begin{aligned} E_1 : \mathbf{e_1} &= \begin{bmatrix} s_1 & 0 & h_1 \end{bmatrix}^T \\ E_2 : \mathbf{e_2} &= \begin{bmatrix} -s_2 \sin\frac{\pi}{6} & s_2 \cos\frac{\pi}{6} & h_2 \end{bmatrix}^T \\ E_3 : \mathbf{e_3} &= \begin{bmatrix} -s_3 \sin\frac{\pi}{6} & -s_3 \cos\frac{\pi}{6} & h_3 \end{bmatrix}^T \end{aligned} \tag{15} $$

Taking the average of the vertices of the top plate relates the end effector position, $P : \mathbf{p} = [x, y, z]^T$, to the internal variables as follows:

$$ \begin{aligned} x &= \frac{1}{3}\left(s_1 - \frac{(s_2 + s_3)}{2}\right) \\ y &= \tfrac{\sqrt{3}}{6}(s_2 - s_3) \\ z &= \tfrac{1}{3}(h_1 + h_2 + h_3) \end{aligned} \tag{16} $$

Solving (13), (14), and (16) simultaneously for s_i and h_i, $i \in (1, 2, 3)$ then substituting into (10) gives the following expressions for each of the three scissor lengths:

$$ L_1 = \sqrt{\begin{aligned} &\left(-\frac{L_t}{\sqrt{3}} + r + s_0 - 2x\right)^2 \\ &+ \left(\sqrt{\frac{2}{3}\left(\sqrt{3}L_t + 3x\right)(r - x) - 2y^2} - z\right)^2 \end{aligned}} \tag{17} $$

$$ L_2 = \sqrt{\begin{aligned} &\left(-\frac{L_t + 3y}{\sqrt{3}} + r + s_0 + x\right)^2 \\ &+ \frac{\left(-3rT_2 - 3T_2 x + \sqrt{6}T_1 y + 6yz\right)^2}{36y^2} \end{aligned}} \tag{18} $$

$$ L_3 = \sqrt{\begin{aligned} &\left(-\frac{L_t - 3y}{\sqrt{3}} + r + s_0 + x\right)^2 \\ &+ \frac{\left(3rT_2 + 3T_2 x + \sqrt{6}T_1 y + 6yz\right)^2}{36y^2} \end{aligned}} \tag{19} $$

where,

$$ r = \sqrt{x^2 + y^2} \tag{20} $$

$$ T_1 = \sqrt{\left(\sqrt{3}L_t + 3x\right)(r - x) - 3y^2} \tag{21} $$

$$ T_2 = \sqrt{2\left(\sqrt{3}L_t + 3x\right)(r - x) - 6y^2} \tag{22} $$

(17), (18), and (19) can each be substituted in to (4) to give the full inverse kinematics for a segment of the expansion arm.

C. Orientation Expressions

The final kinematic expression necessary for implementing the Sequential Expansion Algorithm is the normal vector, $\mathbf{b}$ of the top plate as a function of top plate position, $p = [x, y, z]$.

First we obtain expressions for the coordinates of the spherical joints, E_1, E_2, and E_3, in terms of the top plate position, P, by substituting our expressions for the internal variables found above into (15).

Next we can express the unit vectors, $\mathbf{n}$ and $\mathbf{t}$, as functions of the coordinates of the vertices, E_1, E_2, and E_3:

$$ \begin{aligned} \mathbf{n} &= \frac{1}{L_{tv}}(\mathbf{e_1} - \mathbf{p}) \\ \mathbf{t} &= \frac{1}{L_t}(\mathbf{e_2} - \mathbf{e_3}) \end{aligned} \tag{23} $$

In (23), L_{tv} represents the radius of the circumcircle of triangle composed of the vertices, E_i, $i \in (1, 2, 3)$.

Let the components of $\mathbf{b}$ be denoted $[b_X, b_Y, b_Z]$. $\mathbf{b}$ is defined by the cross product, $\mathbf{n} \times \mathbf{t}$. This operation gives the following expressions for the components of $\mathbf{b}$ as functions of the top plate position, P:

$$ b_X = \frac{\sqrt{2\left(\sqrt{3}L_t + 3x\right)(r - x) - 6y^2}}{L_t^2} \tag{24} $$

$$ b_Y = \frac{(r + x)\sqrt{2\left(\sqrt{3}L_t + 3x\right)(r - x) - 6y^2}}{L_t y} \tag{25} $$

$$ b_Z = \frac{L_t - 2\sqrt{3}r}{L_t} \tag{26} $$

D. Compensation for Coupled Sideway Deviation

$\mathbf{b}$ should point toward the next waypoint so the point can be reached via a pure elongation. We can express the coordinates of the next waypoint, $\mathbf{C}$, as follows:

$$ Point\,\mathbf{C} : \begin{bmatrix} C_x \\ C_y \\ C_z \end{bmatrix} = \begin{bmatrix} x + kb_x \\ y + kb_y \\ z + kb_z \end{bmatrix} \tag{27} $$

where k is the distance between points P and B. Note that the coordinates of point $\mathbf{C}$ are known. Also note that the height of the segment being considered is specified: $z = z^*$. Then substituting for the axial extension, $z^* = \mathbf{OA}$, for z, and b_z from (26) we

Authorized licensed use limited to: Worcester Polytechnic Institute. Downloaded on September 02,2026 at 14:14:37 UTC from IEEE Xplore. Restrictions apply.

find:

$$k = \frac{C_z - z^*}{1 - 2\sqrt{3}r/L_t} \tag{28}$$

Substituting (24), (25) and (28) into (27) we obtain two conditions on x and y:

$$C_x = x + \frac{(C_z - z^*)(\sqrt{2\left(\sqrt{3}L_t + 3x\right)(r-x) - 6y^2})}{(1 - 2\sqrt{3}r/L_t)L_t^2}$$

$$C_y = y + \frac{(C_z - z^*)(r+x)\sqrt{2\left(\sqrt{3}L_t + 3x\right)(r-x) - 6y^2}}{(1 - 2\sqrt{3}r/L_t)L_t y} \tag{29}$$

Solving (29) gives the position, p^* which assures that the proximal unextended stage will reach $\mathbf{C}$ when extended distance k.

Due to the non-linear nature of these coupled equations, obtaining a closed form solution is not tractable. This problem can, however, be reformulated and solved by using a simple recursive formula. The cost function, defined in (30), is to quantify the alignment of the unit vector $\mathbf{b}$ given by (24), (25), and (26) the unit vector, $\mathbf{T}$, pointing in the direction of Point C from the current position of point P, as given by (31) below.

$$J = \mathbf{b}^T \mathbf{T} \tag{30}$$

$$\mathbf{T} = \frac{\mathbf{C} - \mathbf{p}}{\|\mathbf{C} - \mathbf{p}\|} \tag{31}$$

The optimal position for the segment is given by:

$$\mathbf{p}^* = \arg\min_{\mathbf{p}}(J) \tag{32}$$

This optimization is solved in real time using gradient descent. The results of the optimization was inserted into the constraint equations, (29) to verify the validity of the solutions.

V. Prototype and Experiment

A prototype of TSERA was built to evaluate the feasibility of the proposed design concept and control algorithm. The initial prototype is pictured in Fig. 9. Each segment can extend to a maximum height of 240 mm with an extension ratio of 5.3. The prototype weighs 560 grams. The effective diameter, measured by the largest circumcircle of the robot, excluding electronics, is 155 mm. This gives a slenderness ratio of 1.5 for each stage. The maximum tilt angle of a single stage is 75 degrees when tilting about an arbitrary axis perpendicular to $\mathbf{Z}$. This metric was determined using the singularity analysis presented in [18].

The actuator assembly is shown in Fig. 8(B). The height of each scissor mechanism is controlled by a single actuator. It features a bi-directional lead screw and two lead nuts. The opposing threading of the two sides of the lead screw creates symmetric linear motion between the two lead nuts. The distance between the two lead nuts determines the height of a single scissor linkage.

The base of a single segment without the scissor linkage is depicted in Fig 8(A). The triangular shape and vertical side walls add stiffness in both radial and axial directions. This component

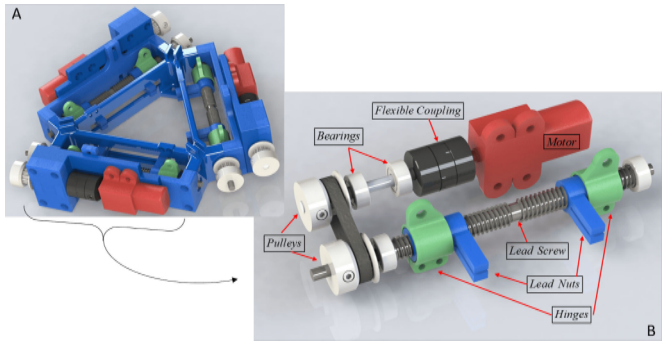


Fig. 8. Base and actuator assembly for a single segment.

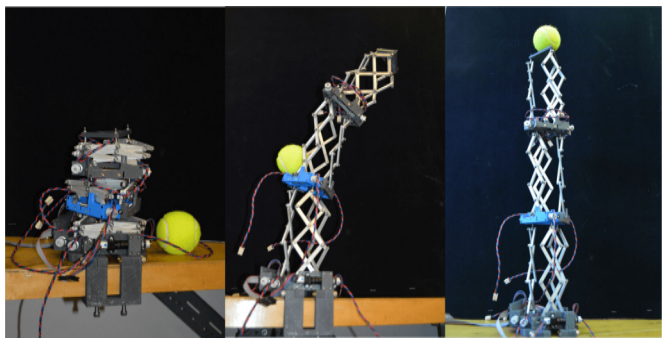


Fig. 9. Various poses of TSERA: Left, fully contracted, Middle, halfway, Right, fully extended. The yellow tennis ball shows the size of the robot.

is hollow down the center to minimize weight and allow central wiring for motor control and sensor integration. The cutout in each of the three side walls acts as a compact linear guide for the lead nuts in the actuator assembly. The linear guide is preloaded in rotation along the axis of travel to prevent backlash with a set of cantilever beams protruding from the lead nuts.

The top plate, base frame, and lead nuts were all 3D printed on a Stratasys Dimension 1200es printer using ABS. The scissor linkages were waterjet cut from 6061 aluminum.

The motion controller for this system was implemented on an Arduino Mega using Cytron Dual Channel Motor Drivers and a 15 V switching power source. Each DC motor drives a lead screw assembly via the timing belt and pulley system depicted in Fig. 8(B). The motor encoders were calibrated at the start of each new trial via optical limit switches.

Simple PID control was implemented on each of the scissor mechanism. The positioning repeatability along the Z-axis of each segment is 80 microns. The axial stiffness of a segment of the prototype is 57 N/mm. Displacement was measured over 30 trials for both repeatability and stiffness using an SPI dial gauge with 10 micron graduations. During stiffness testing, a single segment was loaded with 500, 700, and 900 gram sample masses along the Z-axis while measuring displacement with the SPI dial gauge.

Fig. 10 shows prototype TSERA elongating through a confined space. The robot followed the trajectory generated with the Sequential Expansion Algorithm where the coupled sideway deviation was compensated for. Fig. 11 shows the process of computation. The transparent tube represents the collision-free

Authorized licensed use limited to: Worcester Polytechnic Institute. Downloaded on September 02,2026 at 14:14:37 UTC from IEEE Xplore. Restrictions apply.

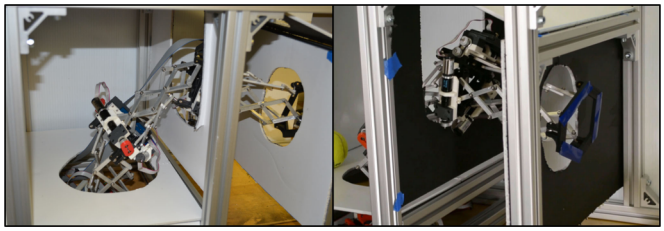


Fig. 10. Multiple views of TSERA navigating a confined space.

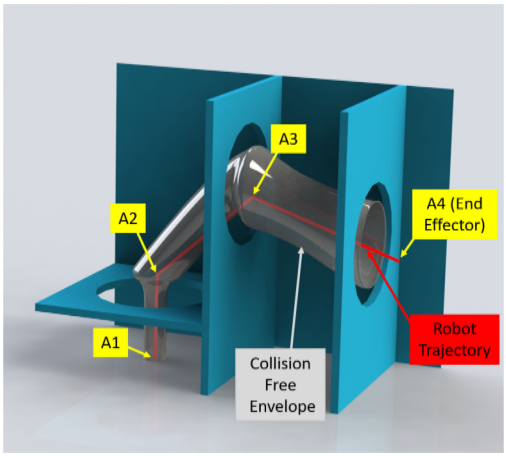


Fig. 11. Rendering of a successful traversal of a simulated confined space using the Sequential Expansion Algorithm as implemented for TSERA.

configuration space. The red line represents the centerline of the robot where each of the vertices are a waypoint or interface between segments of the robot. The output is a list of target position vectors in the reference frame of each of the segments base coordinate frame, O, as depicted in Fig. 7. A set of evenly spaced points are then generated between the current position of each segment and the target position. Driving this trajectory through the inverse kinematics for each stage in sequence provides the reference to the low level motor controllers thus enabling collision free motion.

VI. Conclusion

This letter outlines the design and accompanying motion planning algorithm for a new class of robotic arm capable of accessing confined spaces. Each segment of TSERA is composed of a 3-RPS type zero-torsion parallel mechanism. These segments are outfitted with scissor mechanisms to increase their workspace along their central axis. Concatenating these segments produces a highly redundant manipulator capable of articulating via coordinated and sequential motion between adjacent segments of TSERA.

For typical high DOF systems, solving path planning problems is computationally costly. The segmented structure of TSERA sidesteps this complexity with the Sequential Expansion Algorithm. This approach reduces a high dimensional planning problem to a sequence of three-DOF planning problems, thus largely reducing the computing power required to reach target points in confined environments.

Compared to pneumatically-driven soft robots, TSERA achieved high stiffness and high positioning accuracy. It can also carry an end-effector with significant payload and locate it at an arbitrary position and orientation.

References

[1] W. Felt *et al.*, "An inductance-based sensing system for bellows-driven continuum joints in soft robots," in *Proc. Robot., Sci. Syst.*, 2017.
[2] E. W. Hawkes, L. H. Blumenschein, J. D. Greer, and A. M. Okamura, "A soft robot that navigates its environment through growth," *Sci. Robot*, vol. 2, no. 8, 2017, Art. no. eaan3028. [Online]. Available: http://robotics.sciencemag.org/content/robotics/2/8/eaan3028.full.pdf
[3] Z. M. Hammond, N. S. Usevitch, E. W. Hawkes, and S. Follmer, "Pneumatic reel actuator: Design, modeling, and implementation," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2017, pp. 626–633.
[4] J. D. Greer, T. K. Morimoto, A. M. Okamura, and E. W. Hawkes, "Series Pneumatic Artificial Muscles (sPAMs) and application to a soft continuum robot," in *Proc. IEEE Int. Conf. Robot. Automat.*, May 2017, pp. 5503–5510. [Online]. Available: http://ieeexplore.ieee.org/document/7989648/
[5] F. Matsuno and S. Tadokoro, "Rescue robots and systems in Japan," in *Proc. IEEE Int. Conf. Robot. Biomimetics*, 2004, pp. 12–20.
[6] C. Wright *et al.*, "Design and architecture of the unified modular snake robot," in *Proc. IEEE Int. Conf. Robot. Automat.*, May 2012, pp. 4347–4354.
[7] S. Hirose and H. Yamada, "Snake-like robots: Machine design of biologically inspired robots," *IEEE Robot. Autom. Mag.*, vol. 16, no. 1, pp. 88–98, Mar. 2009.
[8] C. Wright *et al.*, "Design of a modular snake robot," in *Proc. IEEE Int. Conf. Intell. Robots Syst.*, 2007, pp. 2609–2614.
[9] J. M. Friesen *et al.*, "The second generation prototype of a Duct Climbing Tensegrity robot, DuCTTv2," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2016, pp. 2123–2128.
[10] Y. Chen, J. Liang, and I. W. Hunter, "Modular continuum robotic endoscope design and path planning," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2014, pp. 5393–5400.
[11] W. Grundfest, A. Slatkin, J. Burdick, and W. Grundfest, "The development of a robotic endoscope," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. Human Robot Interact. Cooperative Robots*, vol. 2, 1995, pp. 162–171.
[12] T. Lozano-Perez *et al.*, "Spatial planning: A configuration space approach," *IEEE Trans. Comput.*, vol. C-32, no. 2, pp. 108–120, Feb. 1983. [Online]. Available: http://lis.csail.mit.edu/pubs/tlp/spatial-planning.pdf
[13] J.-C. Latombe, *Robot Motion Planning*. Norwell, MA, USA: Kluwer, 1991.
[14] R. Yañez-valdez, "Analysis and stiffness evaluation of a microparallel kinematic," *DYNA*, vol. 84, no. 201, pp. 224–233, 2017.
[15] Z. Wang, S. Ji, Y. Li, and Y. Wan, "A unified algorithm to determine the reachable and dexterous workspace of parallel manipulators," *Robot. Comput.-Integr. Manuf.*, vol. 26, no. 5, pp. 454–460, 2010.
[16] D. J. Gonzalez and H. H. Asada, "Design and analysis of 6-DOF triple scissor extender robots with applications in aircraft assembly *," *IEEE Robot. Autom. Lett.*, vol. 2, no. 3, pp. 1420–1427, Jul. 2017.
[17] D. J. Gonzalez and H. H. Asada, "Triple scissor extender: A 6-DOF lifting and positioning robot," in *Proc. IEEE Int. Conf. Robot. Automat.*, Jun. 2016, pp. 847–853.
[18] I. A. Bonev, "Direct kinematics of zero-torsion parallel mechanisms," in *Proc. IEEE Int. Conf. Robot. Automat.*, May 2008, pp. 3851–3856. [Online]. Available: http://ieeexplore.ieee.org/document/4543802/
[19] S. Briot and I. A. Bonev, "Singularity analysis of zero-torsion parallel mechanisms," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2008, pp. 1952–1957.
[20] F. Collins and M. Yim, "Design of a spherical robot arm with the Spiral Zipper prismatic joint," in *Proc. IEEE Int. Conf. Robot. Automat.*, Jun. 2016, vol. 2016, pp. 2137–2143.
[21] G. Woodruff, P. Muench, and G. Witus, "Zipper mast for enhanced communications and surveillance," *Proc. SPIE*, vol. 8045, May 2011, Art. no. 804512.

Authorized licensed use limited to: Worcester Polytechnic Institute. Downloaded on September 02,2026 at 14:14:37 UTC from IEEE Xplore. Restrictions apply.